# BA ZC581 :
# Applied Data Science
# Module: 15

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# Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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# Module 15:
# Data Science Use cases in Business

# Module 15 Topics

Topic 1: Quick Introduction to Machine Learning

Topic 2: Non ML examples from Finance

Topic 3: Examples from HR, Operations

Topic 4: Examples from Marketing

Ref: T1 , Ch 17, 18

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# Topic 1: Quick Introduction to Machine Learning

# Machine Learning

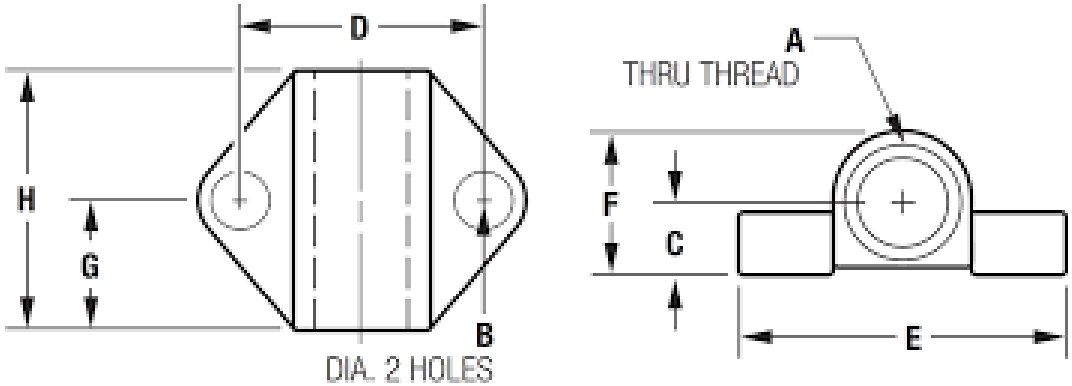
- Machine Learning (ML) is a subset of Artificial Intelligence that enables systems to learn patterns from data and improve performance without being explicitly programmed.
- Machine Learning transforms data into predictive and decision-making intelligence, forming the foundation of modern analytics, automation, and AI-driven systems.
- Example: Instead of programming rules to detect spam emails, ML
  - Learns from thousands of labeled emails
  - Identifies patterns that mostly represent spam
  - Predicts whether new emails are spam

# Machine Learning Model

- Model: A mathematical representation of patterns learned from training data.
- Built through a training process where the algorithm adjusts parameters to minimize error.
- Once trained, the model can predict outcomes for new, unseen data.

# Case Study: Component Testing

Consider the following component




- There are 8 parameters (attributes or dimensions) which should be measured and verified to lie within design limits

Outcome: **REJECT**, **ACCEPT**, **REWORK**

# Case Study: Component Testing

Assume out of these 8, only 2 parameters are important (Principal Components) ex: H and D

Assume design limits are as follows

**ACCEPT:**

H: 75mm ± 1% and

D: 50mm ± 1%

**REWORK:**

H: greater than 75.75mm

D: greater than 50.5mm

**REJECT:** Otherwise

**As per data give by Design Department**

# Rule Based

- **Designer / Expert** devised the **rule(s)** which define the system behaviour (classification criteria)
- Rule Base:

```
If  (74.25 <= H <= 75.75) && ( 49.5 <= D
        <= 50.5) then outcome =ACCEPT
else if (H>75.75 || D> 50.5)
        then outcome = REWORK
        else outcome = REJECT
```

# Component Testing: Model Based

## An Alternate Approach:

| | | | |
|---|---|---|---|
| Assume an expert measures the principal parameters and decides the ‘class’. | These decisions of the expert is recorded and is used to understand the expert’s working (called system behaviour / model) | Use this model to check the outcomes on new data | Validate these outcomes with an expert and refine behaviour |

# Model Based Approach

- Examine the training data and prepare a decision model.
- Use the model to identify the outcome for new unseen data

Seen data- training data

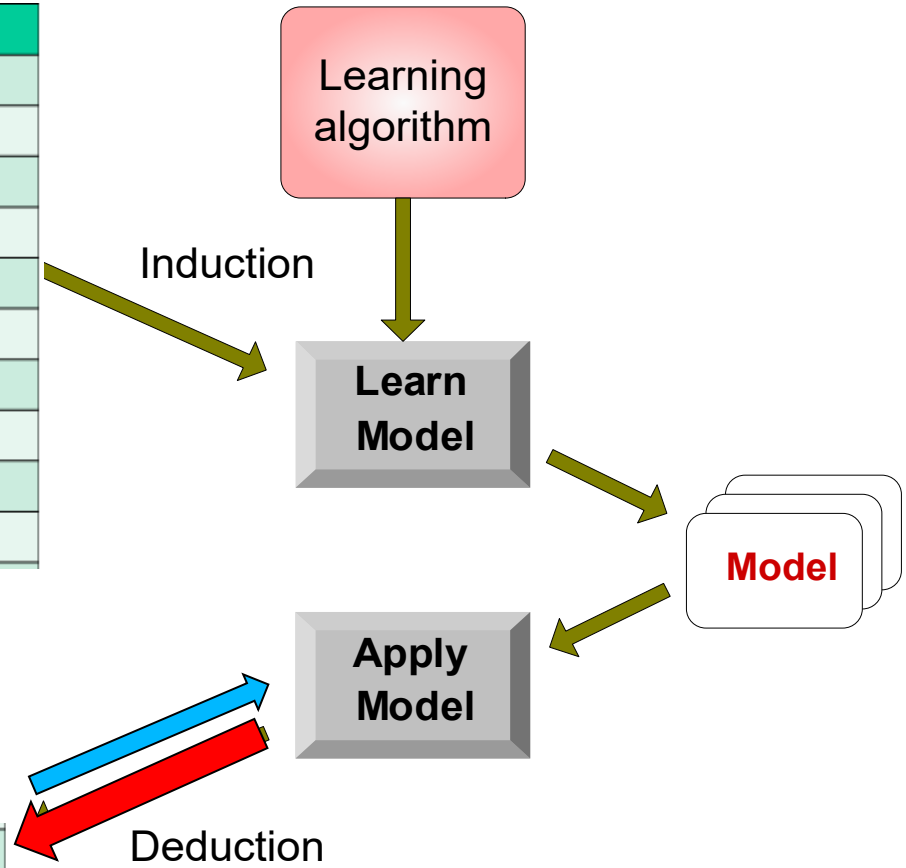
| Item Id | H | D | Outcome |
|---|---|---|---|
| 394 | 74.29 | 49.21 | REEJECT |
| 395 | 75.12 | 50.50 | REWORK |
| 396 | 74.55 | 50.12 | REWORK |
| 397 | 74.99 | 50.22 | REJECT |
| 398 | 74.56 | 50.10 | ACCEPT |
| 399 | 75.82 | 50.11 | ACCEPT |
| 400 | 74.10 | 50.05 | ACCEPT |
| 401 | 77.1 | 51.05 | REJECT |
| 401 | 74.01 | 50.05 | REJECT |
| 402 | 75.0 | 50.22 | ACCEPT |
| .. | | | |
| .. | | | |
| 2571 | 75.82 | 50.22 | ? |
| 2752 | 74.99 | 49.76 | ? |

Unseen data

# Model Based Approach

| Item Id | H | D | Outcome |
|---|---|---|---|
| 394 | 74.29 | 49.21 | REEJECT |
| 395 | 75.12 | 50.50 | REWORK |
| 396 | 74.55 | 50.12 | REWORK |
| 397 | 74.99 | 50.22 | REJECT |
| 398 | 74.56 | 50.10 | ACCEPT |
| 399 | 75.82 | 50.11 | ACCEPT |
| 400 | 74.10 | 50.05 | ACCEPT |
| 401 | 77.1 | 51.05 | REJECT |
| 401 | 74.01 | 50.05 | REJECT |
| 402 | 75.0 | 50.22 | ACCEPT |

Training Set




| | | | |
|---|---|---|---|
| 2571 | 75.82 | 50.22 | ? |
| 2752 | 74.99 | 49.76 | ? |

Test Set

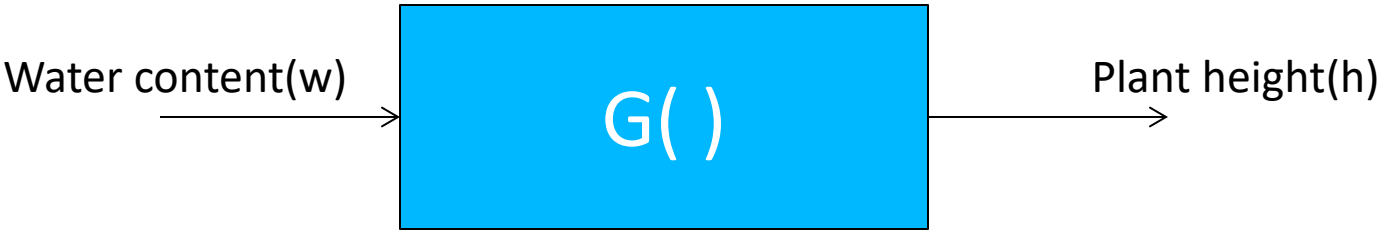
# Data Analysis Uses Models

## Model:

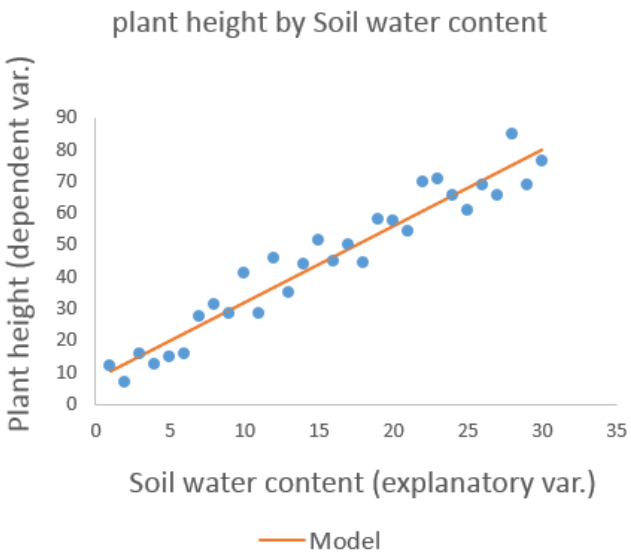
- Defines the behaviour of a phenomenon
- Identifies a mathematical representation of the phenomenon's cause – effect relationship
- Enables us to understand the process / phenomena that generated the data
- Establishes the cause and effect relationships

# Model
# ex: growth model of a plant

Representation of a certain phenomenon




Capture observations and algorithm prepares the model




Linear model

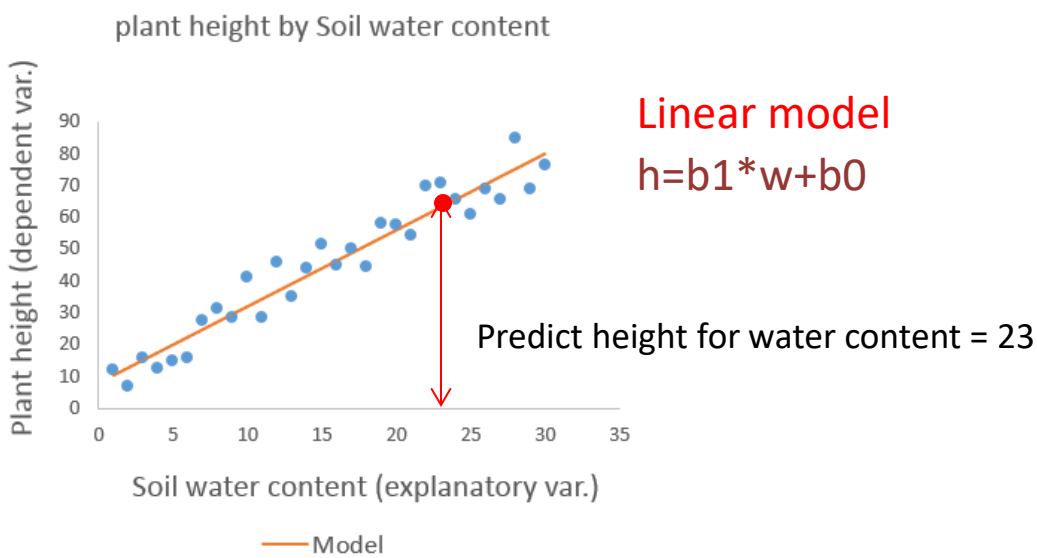
h=b1*w+b0

# Model
# ex: growth model of a plant

## Representation of a certain phenomenon

Water content (w) → h=G(w)=b1*w+b0 → Plant height (h)

## Use the model to predict



# Model Building

Stages

- Collect Data
- Prepare Data (Clean and enrich)
- Create a model (fit the data to a model)
  - May involve training (fit the data many times)
- Test the model
- Refine the model
- Deploy the model for prediction

# Types of Machine Learning

**Supervised Learning**

Learns from labeled data, data which is already classified.

Used for:

Classification (spam vs not spam)

Regression (predict house price)

Example: Given a loan applicant's features, predict the credit risk for loan sanction

**Unsupervised Learning**

Data has no labeled outputs. Finds hidden patterns or affinity to determine labels

Used for : Clustering

Example: Customer segmentation: A bank clusters customers based on spending and income patterns to design targeted loan or credit-card offers

**Reinforcement Learning**

Learns through rewards and penalties

Used in robotics, gaming, recommendation systems

Example: An online retailer tests different prices and learns which ones boost sales by rewarding successful price choices and discouraging poor ones

# Machine Learning Lifecycle

| | |
|---|---|
| **1. Problem Definition** | What decision or prediction are we trying to make?<br>Define success metrics (accuracy, precision, ROI) |
| **2. Data Collection & Preparation** | Gather relevant data<br>Clean, transform, and handle missing values |
| **3. Model Selection & Training** | Choose algorithm<br>Train model on historical data (training data) |
| **4. Model Evaluation** | Validate using test data (validation data)<br>Measure performance (Accuracy, F1-score, RMSE) |
| **5. Deployment & Monitoring** | Deploy into production to work on unseen data<br>Monitor for performance drift<br>Retrain periodically |

# Regression Example

- A supervised learning method used to predict a numeric outcome.
- Models the relationship between a dependent variable (y) and one or more independent variables (X).
- Fits a straight line through data to capture the trend.
- Core equation:

  $y=mX+c$

  here m = slope (impact of X on y), c = intercept.
- Used for forecasting sales, predicting demand, estimating costs, and trend analysis.

# Regression Example

**Training**: The algorithm finds the best-fit line by minimizing prediction error (often using least squares).

**Prediction**: For new X values, the model estimates Y using the learned line.

**Assumptions**:

- Linear relationship between variables
- Independent predictors
- Minimal multicollinearity
- Errors are normally distributed

**Strengths**: Simple, interpretable, fast, widely used in business analytics

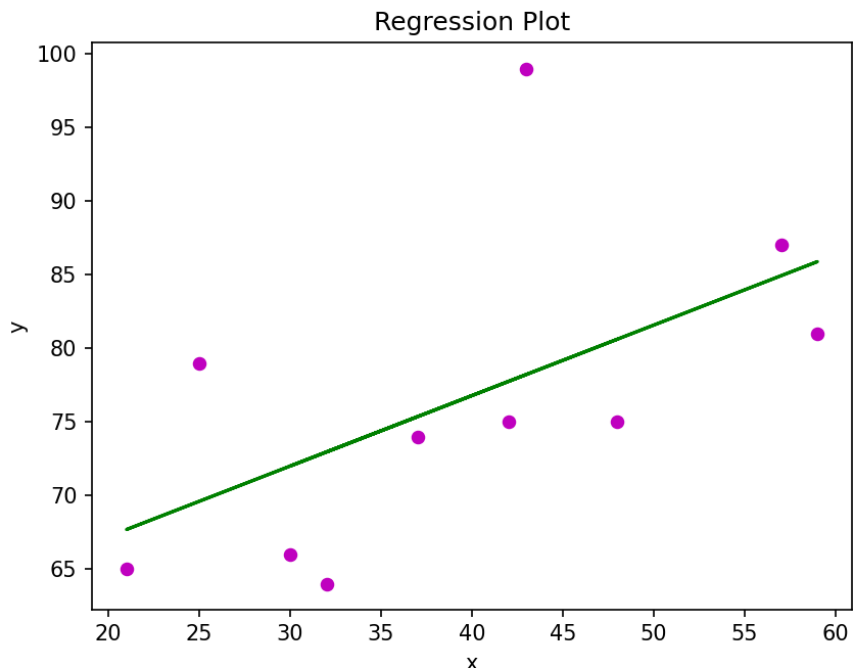
# Linear Regression Example

Scenario: Predict glucose level y) based on age (x)

Training Data:

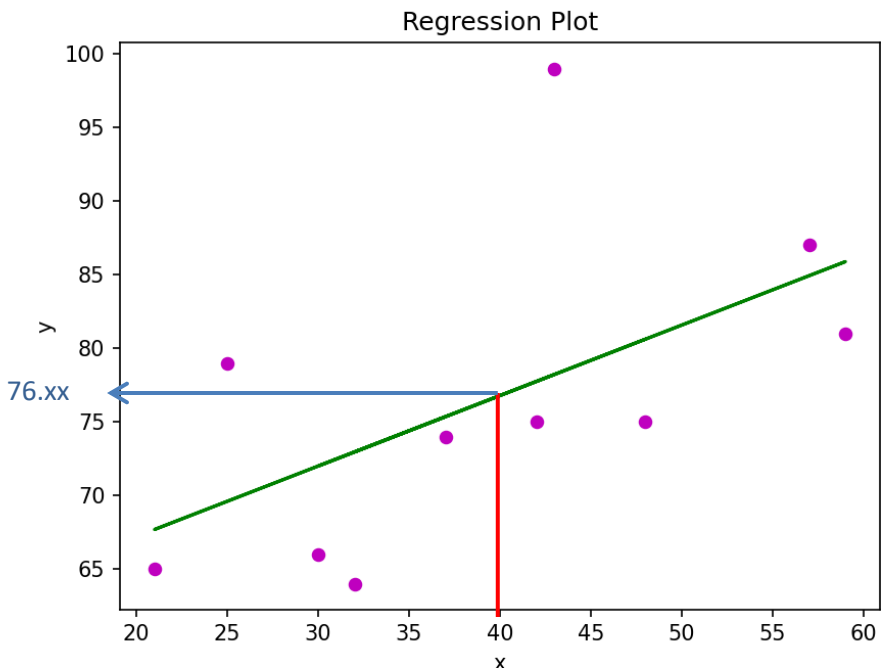
x = np.array([43,21,25,42,57,59,30,37,32,48])

y = np.array([99,65,79,75,87,81,66,74,64,75])



# Use hi model to predict

Say, we wish to predict the glucose level for age = 40




Predicted value = 76.xx

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# Topic 2: Non ML Examples from Finance

# Non-ML Examples from Finance

Non-ML examples of analytics in finance typically involve:

- Descriptive reporting
- Rule-based compliance
- Deterministic financial modeling
- Statistical summaries

# Budget Variance Analysis (Descriptive & Diagnostic Analytics)

- A rule-based comparison of **actual vs budgeted financial performance**.
- **How it works:**
  - Compare planned revenue, costs, and margins with actuals
  - Calculate variance percentages
  - Investigate reasons for deviations
- **Example:**
  - Budgeted Revenue: ₹10 crore
  - Actual Revenue: ₹9.2 crore
  - Variance: -8%
- Finance teams analyze:
  - Price changes
  - Volume shortfalls
  - Cost overruns

# Financial Ratio Analysis (Traditional Statistical Analytics)

- Evaluation of company performance using financial ratios.
- **Common Ratios:**
  - Current Ratio
  - Debt-to-Equity Ratio
  - Return on Investment (ROI)
  - Net Profit Margin
- Example:
  - Return on Investment (ROI) = $\frac{\text{Net Profit}}{\text{Investment}} \times 100$
- Used for:
  - Credit evaluation
  - Investment analysis
  - Performance benchmarking

| Ratio | Interpretation |
|---|---|
| Current Ratio | Liquidity position (≥1 is generally healthy) |
| Debt-to-Equity | Financial leverage level |
| ROI | Investment efficiency |
| Net Profit Margin | Profitability efficiency |

# Rule-Based Fraud Detection (Threshold-Based Controls)

- Fraud detection using fixed business rules instead of predictive models.
- **Example Rules:**
  - Flag transactions above ₹5 lakh
  - Flag transactions from blacklisted countries
  - Alert if 5 failed login attempts occur
- Used in:
  - Banking transaction monitoring
  - AML (Anti-Money Laundering) systems
  - Compliance reporting

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# Topic 3: Examples from HR and Operations

# Employee Turnover & Attrition Analysis (Descriptive Analytics)

- Tracking and analyzing employee exits over a period.

$$\text{Attrition Rate} = \frac{\text{Number of Employees Left}}{\text{Average Number of Employees}} \times 100$$

## Example:

- Employees at start: 200
- Employees at end: 180
- Employees who left: 20
- Attrition rate = 10%

## Business Use:

- Workforce stability monitoring
- HR planning
- Cost of replacement estimation

# Absenteeism Analysis (Rule-Based Monitoring)

- Tracking employee attendance and identifying high absenteeism.

$$\text{Absenteeism Rate} = \frac{\text{Total Absent Days}}{\text{Total Workdays Available}} \times 100$$

**Example:**

- If an employee was absent 6 days out of 60 workdays → 10% absenteeism.

**Typical Rule:**

- Flag employees with absenteeism > 8%

**Business Use:**

- Productivity analysis
- Policy enforcement
- Early HR intervention

# Compensation Ratio (Compa-Ratio) Analysis

- Measures how an employee's salary compares to the midpoint of a salary band

$$\text{Compa-Ratio} = \frac{\text{Employee Salary}}{\text{Salary Band Midpoint}}$$

**Example:**

- Employee salary = ₹9,00,000
- Midpoint = ₹10,00,000
- Compa-ratio = 0.90 (90%)

**Business Use:**

- Pay equity analysis
- Compensation planning
- Promotion and increment decisions

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# Operations ..

# Inventory Reorder Point Analysis (Rule-Based Optimization)

- Determines **when to reorder inventory** using deterministic formulas based on:
  - Average daily demand
  - Lead time
  - Safety stock
- Reorder Point (ROP)=(Average Daily Demand×Lead Time)+Safety Stock

A manufacturing plant data:
Avg daily demand = 200 units
Lead time = 5 days
Safety stock = 300 units

$$ROP = (200 \times 5) + 300 = 1300 \ units$$

- Business impact:
  - Reduces stockouts
  - Controls working capital
  - Improves service levels

# Capacity Utilization & OEE (Overall Equipment Effectiveness)

- Measures how effectively manufacturing equipment is used.
- OEE=Availability × Performance × Quality
- where:
  - Availability = Operating Time / Planned Production Time
  - Performance = Actual Output / Standard Output
  - Quality = Good Units / Total Units

Given,
Availability = 90%
Performance = 85%
Quality = 95%

$$OEE = 0.90 \times 0.85 \times 0.95 = 72.7\%$$

- Business Impact:
  - Identifies bottlenecks
  - Detects machine inefficiencies
  - Improves throughput

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# Topic 4: Examples from Marketing

# Customer Segmentation using RFM Analysis

- Segments customers based on purchasing behavior using three measurable factors:
  - **Recency (R)** – How recently a customer purchased
  - **Frequency (F)** – How often they purchase
  - **Monetary Value (M)** – How much they spend

Customers are scored (e.g., 1–5 scale) on each dimension using quantiles or thresholds.

Example:

| Customer | Recency | Frequency | Monetary | RFM Score |
|---|---|---|---|---|
| A | 5 | 5 | 4 | 554 |
| B | 2 | 3 | 2 | 232 |

Segments can be labeled:

- 555 → "Champions"
- High R, Low F → "New Customers"
- Low R, High F → "At Risk"

Business goal:
Targeted promotions
Loyalty campaigns
Reactivation strategies

# Campaign ROI & Conversion Funnel Analysis

- Evaluates marketing campaign performance using deterministic calculations. Key metrics are:

**Conversion Rate**

$$\text{Conversion Rate} = \frac{\text{Conversions}}{\text{Total Visitors}} \times 100$$

**Cost Per Acquisition (CPA)**

$$CPA = \frac{\text{Campaign Cost}}{\text{Conversions}}$$

**Marketing ROI**

$$ROI = \frac{\text{Revenue} - \text{Campaign Cost}}{\text{Campaign Cost}} \times 100$$

Example:

- Visitors = 10,000
- Conversions = 500
- Campaign Cost = ₹200,000
- Revenue = ₹600,000

Conversion Rate = 5%

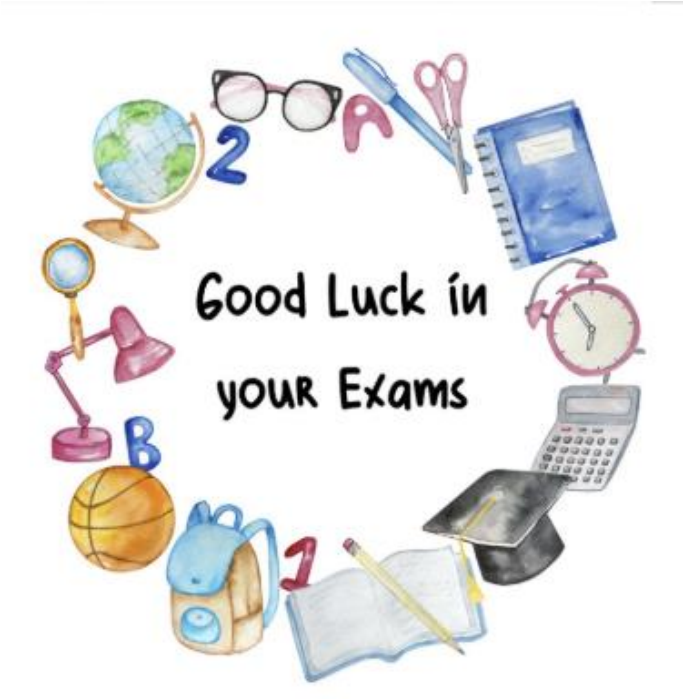
CPA = ₹400

ROI = 200%

Marketing impact:
Budget allocation decisions
Channel performance comparison
Funnel drop-off analysis

# Summary

- Studied the concept of Models and the Machine Learning framework
- Understood the Regression model
- Explored a few non ML models in Finance, HR, Operations and Marketing

# Questions?

# Good Luck